Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Knowledge organiser

Learning goal

I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation.

Calculate a reciprocal

Relative rate = 1/time. For 120 s: 1/120 = 0.00833 s^{-1}. For 90 s: 1/90 = 0.0111 s^{-1}. A shorter endpoint time gives a larger reciprocal, for matched amounts and the same stopping rule.

Describe then qualify

These illustrative first-series points rise from pH 4 to 7 and fall at pH 8. pH 7 is highest among the values tested here. A separate pH 7 repeat gives 120 s (0.00833 s^{-1}); it shows variation, not a new pH value.

Explain and evaluate

pH can change how a substrate binds and how the enzyme speeds up the change. The graph does not show the active-site shape itself. It does not prove permanent denaturation. Match conditions to make a fair comparison. Samples taken 30 s apart limit how precise the time is. A shorter fixed gap narrows that window. It must be feasible and authorised.

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Transfer the science

Enzyme digestion changes molecules in a chemical way. It breaks large molecules into smaller ones. Diffusion is net movement of particles. They move from higher to lower concentration. That movement does not itself break starch into sugars in a chemical way.

Diagram

Dark dots show the first-series rates. The graph uses the supplied data; it is not a record of your own practical.

